# Auto Ticket Classification using Flow Designer

**Team ID: SWTID-2026-2949**

## Project Description:

Auto Ticket Classification harnesses the low-code automation of ServiceNow Flow Designer to classify and route support tickets automatically. Whenever a new incident is created, the flow reads the short description and description, identifies the issue type, and sets the Category, Subcategory, Assignment Group and Priority without any manual triage. The solution consists of three parts: a Ticket Classifier, which maps keywords in the ticket to the right category; an Auto-Assignment step, which routes the ticket to the correct support group; and a Priority Setter, which assigns a priority based on urgency and impact keywords.

Built on a ServiceNow Personal Developer Instance (PDI), the flow runs inside the platform, so no external servers or code deployment are needed. Roles and access controls keep the process secure, and the flow adds a work note to every ticket so that each automated decision can be audited. This reduces triage time, removes misrouted tickets and lets support teams focus on resolving issues instead of sorting them.

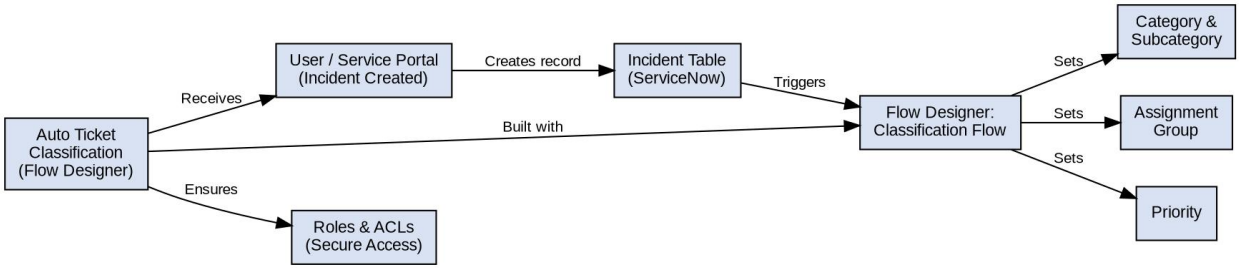
## Scenarios

**Scenario 1:** An employee raises an incident with the short description "Unable to connect to VPN from home". The flow detects the keyword "VPN", sets the category to Network, assigns the ticket to the Network Support group and sets the priority to Moderate, all within seconds of creation.

**Scenario 2:** A user submits "Laptop screen is broken and not turning on". The flow classifies the ticket as Hardware, routes it to the Hardware Support group and adds a work note that records the keyword that triggered the rule.

**Scenario 3:** A ticket arrives with "Production database is down, all users affected". The flow detects the urgent keywords, classifies it as Database, assigns it to the Database Admin group and sets the priority to Critical so that it is picked up immediately.

# Technical Architecture:




**Description:** The solution follows a simple three-layer design on the ServiceNow platform. The input layer is the Incident table, which receives tickets from the Service Portal, email or the platform UI. The automation layer is a Flow Designer flow that is triggered when an incident is created; it evaluates the ticket text with If conditions, looks up the matching assignment group and updates the record. The data layer stores the updated incident, with roles and ACLs controlling who can view or change the flow and the ticket fields. Because everything runs inside the instance, no external hosting is required.

**(Note: This project uses ServiceNow's built-in Incident table and standard tables only, so no separate ER diagram is required.)**

# Pre-requisites:

- **ServiceNow Platform Knowledge:** Basic navigation, tables, forms and lists in ServiceNow.
- **Flow Designer Familiarity:** Triggers, actions, flow logic and data pills.
- **ServiceNow Developer Account:** Free account to request a Personal Developer Instance (PDI).
- **ITSM Basics:** Incident management, categories, priorities and assignment groups.
- **User and Group Administration:** Creating users, groups and roles.
- **Testing Tools:** Browser, Postman and Apache JMeter for functional and performance testing.

# Project Workflow:

**Milestone 1: Platform Setup and Architecture**

- **Activity 1.1:** Create a ServiceNow Personal Developer Instance.
- **Activity 1.2:** Define the classification requirements, categories and routing rules.
- **Activity 1.3:** Define the architecture of the solution.
- **Activity 1.4:** Set up the instance with users, groups and roles.

**Milestone 2: Core Functionalities Development**

- **Activity 2.1:** Develop the core functions: Ticket Classifier, Auto-Assignment and Priority Setter.
- **Activity 2.2:** Configure the incident form and fields used by the flow.

**Milestone 3: Flow Designer Development**

- **Activity 3.1:** Build the Auto Ticket Classification flow with trigger, conditions and update actions.

**Milestone 4: Service Portal and Notifications**

- **Activity 4.1:** Create a record producer in the Service Portal for raising tickets.
- **Activity 4.2:** Add work notes and email notifications to show the result of the classification.

**Milestone 5: Testing and Deployment**

- **Activity 5.1:** Test the flow with sample tickets and review the flow execution.
- **Activity 5.2:** Run performance tests and activate the flow, moving it with an update set.

**Milestone 6: Conclusion**

# Milestone 1: Platform Setup and Architecture

In this milestone, we prepare the ServiceNow environment and decide how tickets will be classified. This involves getting an instance, studying the ITSM categories and support teams, and designing a simple architecture that connects the incident, the flow and the assignment groups.

## Activity 1.1: Create a ServiceNow Personal Developer Instance

Before building the flow, you need a ServiceNow instance where Flow Designer is available. Follow the steps below.

- **Step 1 — Create a Developer Account:**

Visit https://developer.servicenow.com and sign up for a free account.

[Insert Screenshot: ServiceNow Developer sign-up page]

- **Step 2 — Request an Instance:** After logging in, click Request Instance and choose the latest available release. Wait until the instance is ready.

[Insert Screenshot: Request Instance / instance ready]

- **Step 3 — Log in as Administrator:** Open the instance URL and log in with the admin credentials shown in the developer portal. Keep the credentials safe and do not share them.

[Insert Screenshot: ServiceNow instance home page]

- **Step 4 — Open Flow Designer:** In the application navigator, search for "Flow Designer" (Workflow Studio) and open it to confirm it is available.
- **Important —** Personal developer instances go to sleep after a period of inactivity, so log in regularly to keep the instance active.

## Activity 1.2: Define the Classification Requirements

Here are the things that need to be defined before building the flow:

- **Understand the Project Requirements:** Review the support process and identify the tickets that must be classified and routed automatically.
- **Identify the Categories:** Use the standard incident categories Hardware, Software, Network, Database and Inquiry / Help.
- **Define the Keywords:** List the keywords that identify each category and the urgency words that set the priority.

- **Define the Routing Rules:** Map each category to a support group, as shown in the table below.

| Category | Example Keywords | Assignment Group | Default Priority |
|---|---|---|---|
| Network | vpn, wifi, internet, network, router | Network Support | 3 - Moderate |
| Hardware | laptop, screen, keyboard, printer, battery | Hardware Support | 3 - Moderate |
| Software | install, application, crash, license, update | Software Support | 4 - Low |
| Database | database, sql, query, data loss | Database Admin | 2 - High |
| Inquiry / Help | how to, request, access, password reset | Service Desk | 4 - Low |

Urgency keywords such as "down", "outage", "urgent", "all users" and "production" raise the priority to 1 - Critical.

## Activity 1.3: Define the Architecture of the Application

- **Draft an Architectural Diagram:** Create a visual representation showing the Incident table, the Flow Designer flow and the assignment groups.
- **Detail Input Functionality:** Outline how tickets are created through the Service Portal, email or the platform UI.
- **Outline Flow Responsibilities:** Specify that the flow starts when an incident is created, checks the ticket text, finds the matching group and updates the incident.
- **Describe Integration Points:** Define how the flow reads Short description and Description, uses data pills for values, looks up the assignment group and updates the Category, Assignment group and Priority fields.

## Activity 1.4: Set Up the Instance

- **Create Assignment Groups:** Go to User Administration > Groups and create Network Support, Hardware Support, Software Support, Database Admin and Service Desk.

[Insert Screenshot: Assignment groups list]

- **Create Test Users:** Create a few users and add them as members of the groups so that tickets can be assigned and tested.
- **Assign Roles:** Give the group members the itil role so that they can work on incidents.
- **Create an Update Set:** Create a new update set named "Auto Ticket Classification" and make it current so that all changes are captured.

# Milestone 2: Core Functionalities Development

## Activity 2.1: Develop Core Classification Features

Implement the three core features, Classification, Assignment and Priority, as a single flow that runs on every new incident.

- **Ticket Classifier:** Reads the short description and description and sets the Category (and Subcategory where it applies) using keyword conditions. For example, a ticket that contains "VPN" is set to Network.
- **Auto-Assignment:** Looks up the support group for the detected category and sets the Assignment group on the incident.
- **Priority Setter:** Sets the Impact and Urgency values from the keywords so that the platform calculates the Priority, with critical keywords moving the ticket to 1 - Critical.

## Activity 2.2: Configure the Incident Form

- **Show Required Fields:** Add Category, Subcategory, Assignment group, Priority and Work notes to the incident form view.
- **Make Short Description Mandatory:** Ensure the Short description field is mandatory so that the flow always has text to classify.
- **Set a Default Fallback:** Decide that tickets that match no keyword go to the Service Desk group for manual review.

[Insert Screenshot: Incident form with configured fields]

# Milestone 3: Flow Designer Development

Milestone 3 focuses on building the flow that is the backbone of the solution. In this milestone, you will create the trigger, add the classification conditions, update the incident and test each part of the flow.

## Activity 3.1: Building the Auto Ticket Classification Flow

**Create the Flow:**

- Open Flow Designer, click New > Flow and name it "Auto Ticket Classification". Add a short description such as "Automatically classifies and routes new incidents".
- Set the Application to Global and the Run As option to System User so that the flow can update records.

[Insert Screenshot: Create Flow dialog]

**Add the Trigger:**

- Click Add a trigger and choose Created from the Record triggers.
- Select the Incident [incident] table and set the condition Short description is not empty.

[Insert Screenshot: Flow trigger on Incident table]

**Add the Classification Logic:**

- Add an If flow logic block for each category. The first condition checks whether the Short description or Description contains the Network keywords.
- Add an Else If block for Hardware, Software and Database in the same way. An Else block handles tickets that match no keyword.

Example condition for the Network branch:

```
Trigger > Incident Record > Short description   contains   vpn
OR  Short description   contains   wifi
OR  Short description   contains   network
```

**Add the Update Actions:**

- Inside each branch, add a Look Up Record action on the Group [sys_user_group] table to find the matching support group by name.
- Add an Update Record action on the Trigger > Incident Record to set the values below.

```
Category          =  Network
Assignment group  =  Look Up Record > Group Record
Impact / Urgency  =  2 - Medium
```

- Add a final Update Record action in the Else branch that sets the Assignment group to Service Desk.

[Insert Screenshot: Completed flow with If / Else If branches]

**Handle Critical Priority:**

- Add an If block before the category checks that looks for the words "down", "outage", "urgent" and "all users".
- When it matches, update the Impact to 1 - High and the Urgency to 1 - High so that the incident becomes Priority 1 - Critical.

**Add a Work Note:**

- Add an Update Record action at the end of each branch that writes a Work note such as "Auto-classified as Network by flow".
- Click Save and then Activate to enable the flow.

[Insert Screenshot: Flow activated]

# Milestone 4: Service Portal and Notifications

Milestone 4 focuses on giving users an easy way to raise tickets and on making the result of the automation visible. This involves creating a record producer in the Service Portal and adding a notification.

## Activity 4.1: Create a Record Producer

- **Open the Service Catalog:** Go to Service Catalog > Catalog Definition > Maintain Items and click New > Record Producer.
- **Configure the Record Producer:** Name it "Report an Issue", select the Incident table and add the variables Short description and Description.
- **Add to the Portal:** Add the record producer to a catalog category so that it is shown in the Service Portal.

[Insert Screenshot: Record producer "Report an Issue"]

## Activity 4.2: Add Notifications and Work Notes

- **Create an Email Notification:** Go to System Notification > Email > Notifications and create a notification for the Incident table that is sent when the Assignment group changes.
- **Notify the Group:** Send the notification to the assignment group so that the team knows a new ticket has arrived.
- **Review Work Notes:** Open an incident and check that the work note written by the flow shows how the ticket was classified.

[Insert Screenshot: Email notification configuration]

# Milestone 5: Testing and Deployment

In Milestone 5, the focus is on testing the flow with real tickets, checking its performance and then activating it for use. Because ServiceNow runs in the cloud, deployment means activating the flow and moving the update set to other instances when needed.

## Activity 5.1: Functional Testing and Verification

- **Create Test Tickets:** Create incidents from the Service Portal and from the platform UI using the sample descriptions from the scenarios.
- **Check the Result:** Open each incident and verify that the Category, Assignment group and Priority are set correctly.
- **Review the Flow Execution:** Open Flow Designer > Executions and check that each run shows the status Completed.

| S.No | Test Ticket (Short Description) | Expected Category / Group | Expected Priority |
|---|---|---|---|
| 1 | Unable to connect to VPN from home | Network / Network Support | 3 - Moderate |
| 2 | Laptop screen is broken | Hardware / Hardware Support | 3 - Moderate |
| 3 | Application crashes on startup | Software / Software Support | 4 - Low |
| 4 | Production database is down, all users affected | Database / Database Admin | 1 - Critical |
| 5 | Need help with a new request | Inquiry / Service Desk | 4 - Low |

[Insert Screenshot: Incident after auto classification]

[Insert Screenshot: Flow execution details]

## Activity 5.2: Performance Testing and Activation

- **Run Performance Tests:** Use Apache JMeter to send incident creation requests through the REST Table API under load, spike and stress conditions, and note the response time, throughput and error rate.
- **Activate the Flow:** Make sure the flow status is Active and that the trigger condition is correct.
- **Move the Update Set:** Mark the update set as Complete, so that it can be exported or committed to another instance.
- **Important Notes:** Keep the flow simple, avoid unnecessary lookups, and run it asynchronously so that incident creation stays fast.

# Exploring the Application's Pages:

## Incident Form:

[Insert Screenshot: Incident form showing classified fields]

**Description:** The standard ServiceNow incident form where each ticket is created and updated. After the flow runs, the Category, Assignment group and Priority are filled in automatically, and a work note records the decision.

## Flow Designer Page:

[Insert Screenshot: Flow Designer canvas]

**Description:** The visual canvas where the Auto Ticket Classification flow is built. It shows the Created trigger, the If / Else If branches, the Look Up Record actions and the Update Record actions, along with the execution history for debugging.

## Service Portal – Report an Issue:

[Insert Screenshot: Service Portal record producer]

**Description:** The user-facing form in the Service Portal where employees describe their problem. Submitting the form creates an incident, which triggers the flow immediately.

## Conclusion:

Auto Ticket Classification using Flow Designer is a low-code automation solution built on ServiceNow that removes the manual work of sorting support tickets. It uses a Flow Designer flow to read each new incident, set the category, route it to the right assignment group and assign a priority, with a work note that records every decision. The project, which included platform setup, flow development, Service Portal integration and testing, shows how low-code automation can speed up ticket handling, reduce misrouted tickets and give support teams more time to resolve issues. Looking ahead, the solution can be expanded by adding machine learning based classification using Predictive Intelligence, supporting more categories and languages, and adding dashboards to track classification accuracy and resolution times. By continuously refining the keyword rules and monitoring flow performance, the solution can grow into a complete intelligent service desk for teams of all sizes.